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Assignment 3

Q1 Inputs

T = temperature sensor (0 if < 30°c, 1 if ≥ 30°c)
P = Pressure Sensor (0 if < 50 psi, 1 if ≥ 50 psi)
M = Master Alarm switch (1 is On, 0 is Off)

W = warning light (1 = on , 0 = off)

Output

when M = 1, w = 1 and if M = 0, T = 1 and P = 1 than Output(w) = 1

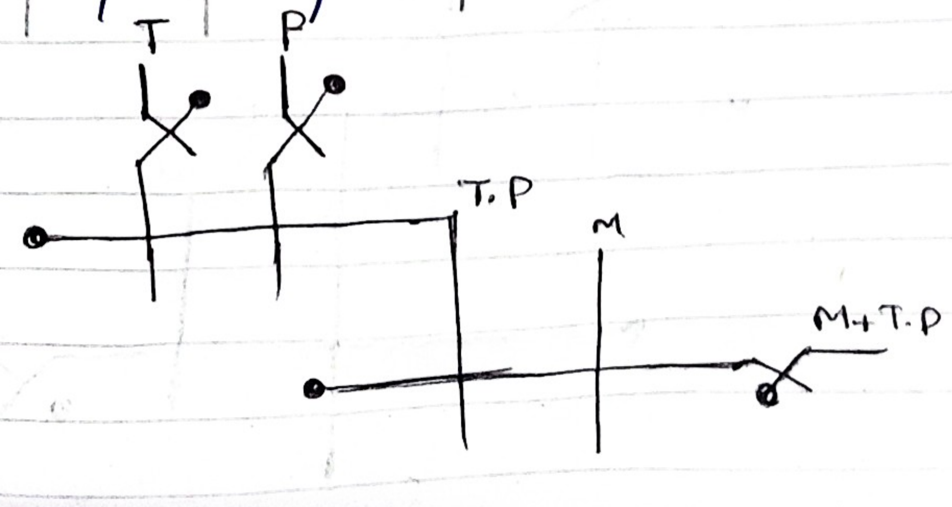
M	T	P	W
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Equation

$$W = M + M'TP$$

Or

$$W = M + T.P$$



Q2 Input

W = water level sensor (1 if >10, 0 if <=10)
F = weather for cast (1 if rain expected 0 not)
B = Backup Battery sensor (1 if full charge
0 if < 20%)

Output

P = Pump (0 or 1)

Output 1 when W = 1 and F = 1 and B = 1.

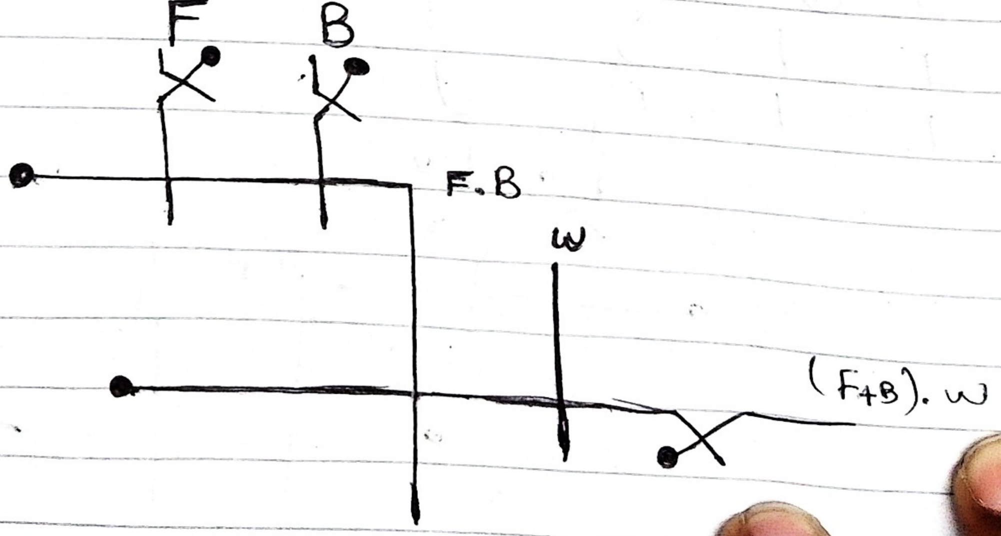
W	F	B	P
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Equation

$$\boxed{P = W + W'FB}$$

or

$$\boxed{P = W + FB}$$



3

T = Thermostat (1 if > 80°C, else 0)

A = Ambient room sensor (1 if > 25°C else 0)

M = Maintenance Switch (1 if activity working else 0)

Outputs

Output 1 when if T = 1 and M = 0 and if T = 1 and A = 1

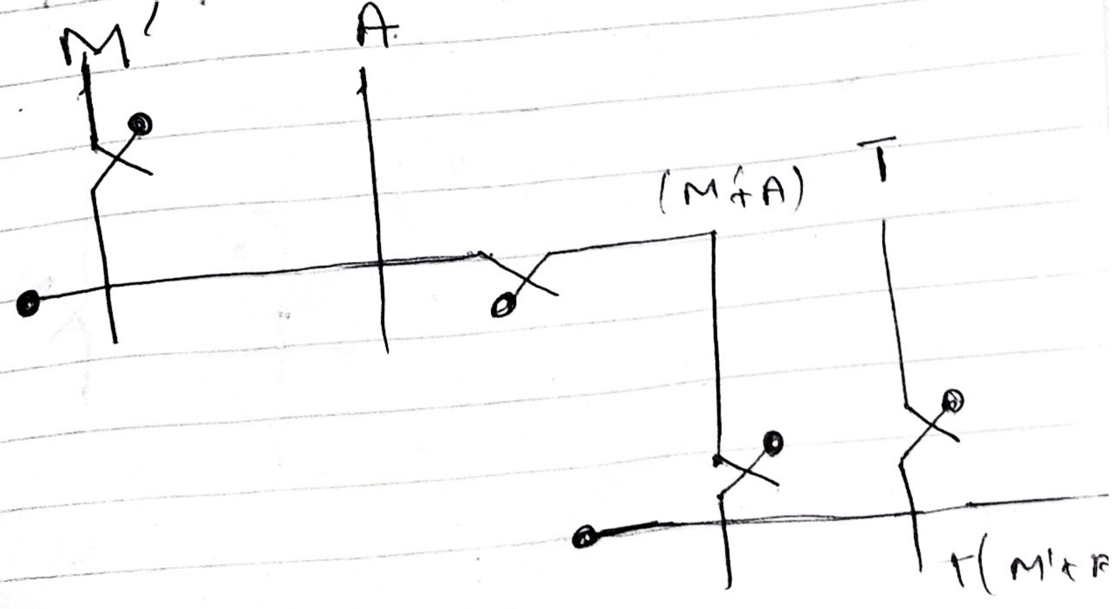
T	A	M	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Equation

"$F = TM' + TA$"

Or

"$F = T(M' + A)$"



Q4

Input

M = Manager's Key (1 when inserted & turned)
C = Cashier's Key (1 when inserted & turned)
T = Time locle (1 when 9 am to 5 am else 0)

Logic

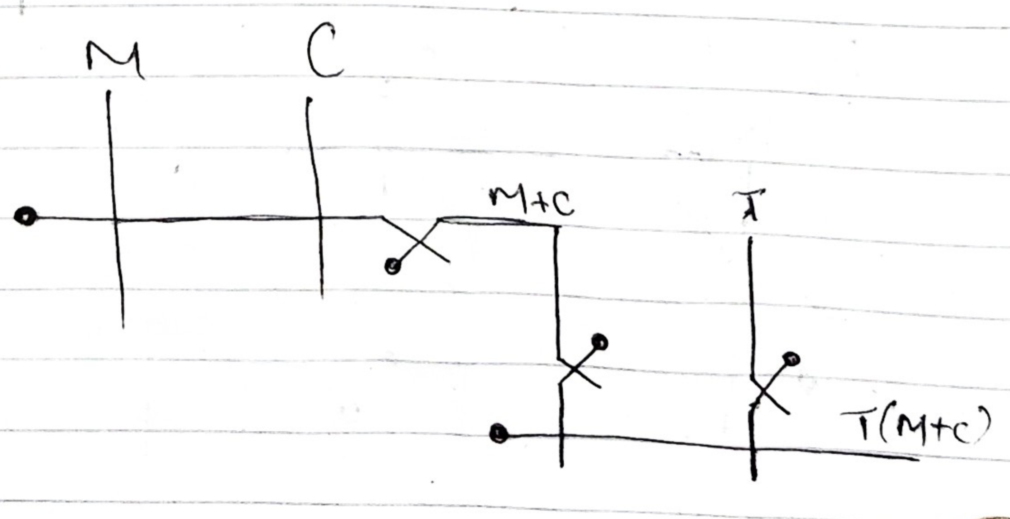
when T = 1 and M = 1 or C = 1] Output 1

V = 0 if T = 0

M	C	T	V
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Equation

$V = T(M+C)$



Q#5 Input

T = Exterior thermostar (1 if < 4°C els

S = Sensor (1 > 70% else 0)

M = Manual button (1 or 1)

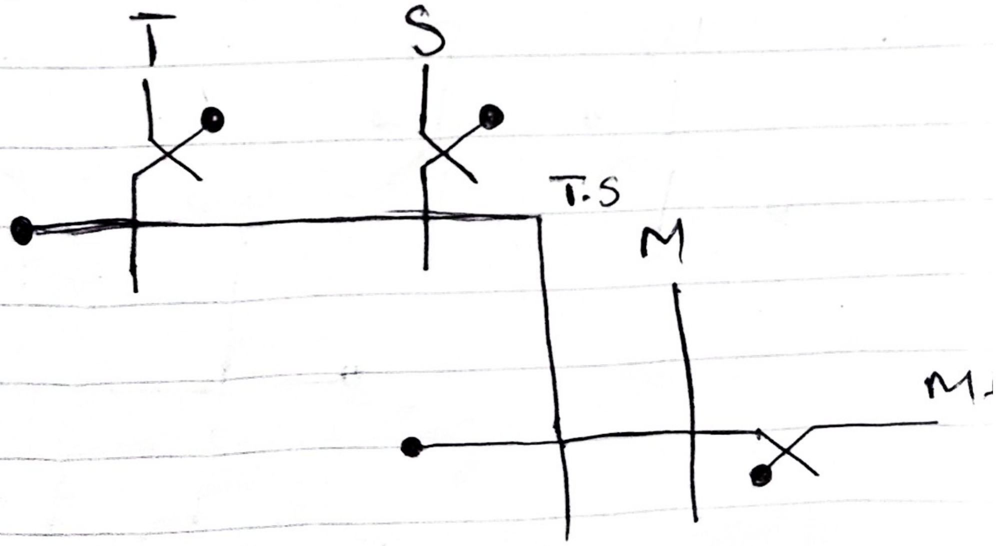
Output

when T = 1 and S = 1, M = 1] Outpu

T	S	M	D
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Equation

$D = T.S + M$



Q6 Input

W = Weight (1 if > 5 kg)

D = Dimension (1 if > 50 cm)

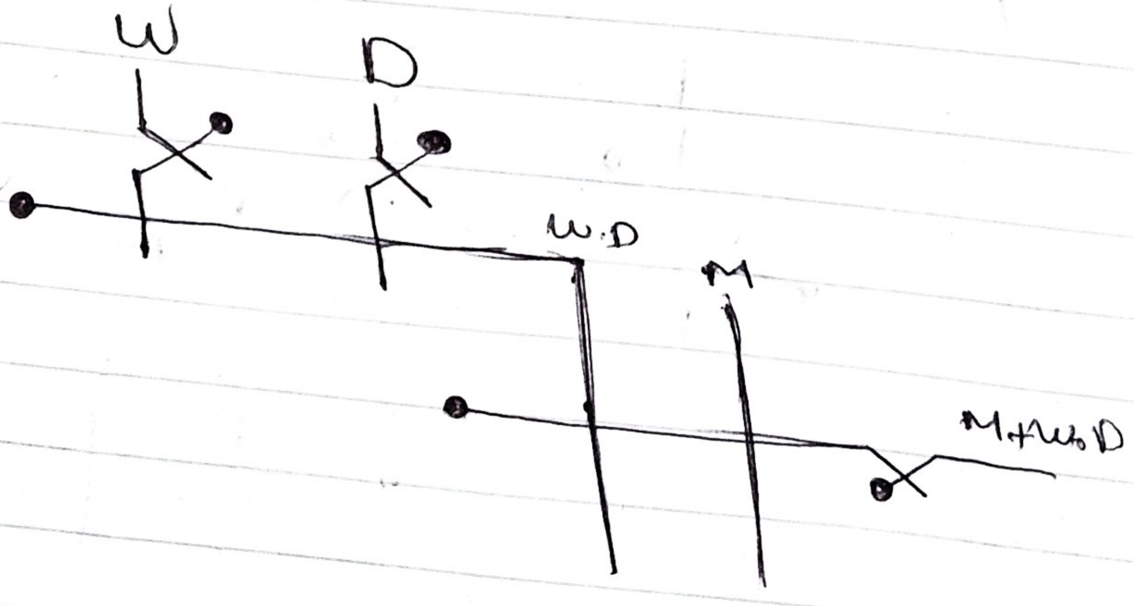
M = Metal detactor (1 if foriegn metal is detected)

when M = 1, W = 1 and D = 1] Output 1

W	D	M	R
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Equation

$R = M + W.D$



Q1 Input

L = Laser tripwire (1 if standing in the doorway)

S = sensor (1 if > 1000 kg)

B = mechanical bumper (1 if physically strikes an Object)

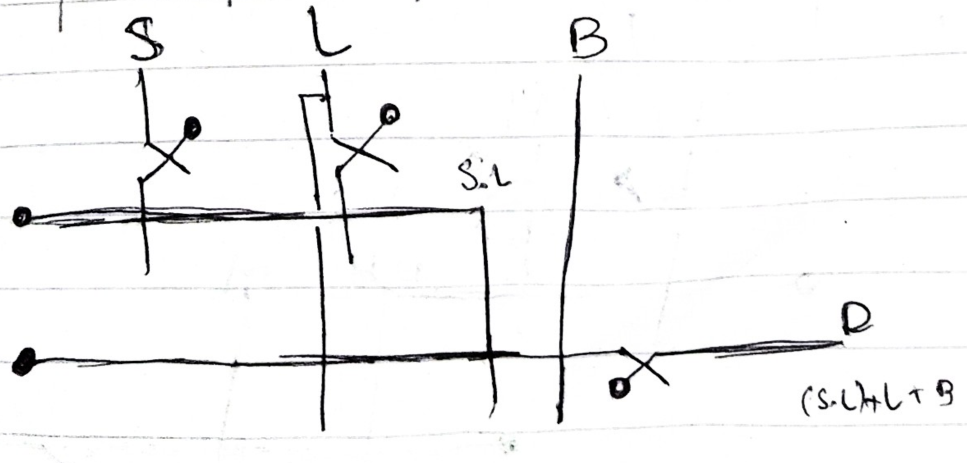
Output

Output = 1 when
B = 1 or L = 1
S = 1 & L = 1

L	S	B	D
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Equation

$D = B + L + (S.L)$



Q8

Input

D = Smoke detector (1 if > 20%)
H = Heat sensor (1 if > 75 °C)
M = Manual pull (1 if pulled by an employee)

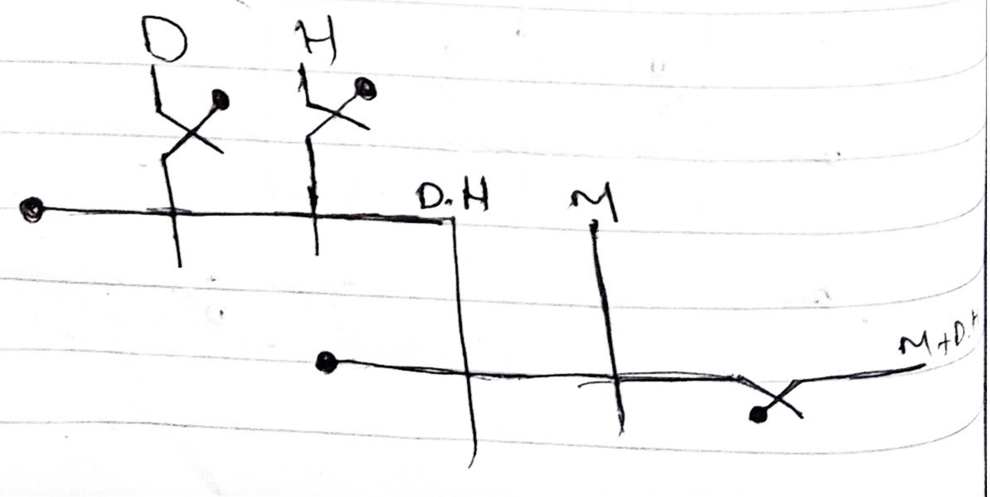
Output

S = Sprinkler (0 or 1)
S = 1 if:
* D = 1 and H = 1
  M = 1

D	H	M	S
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Equation

$S = D \cdot H + M$



Q9

Input

L → Light sensor (1 if > 500 lux, 0 if < 500 lux)
M → Motion sensor (1 if all chickens inside coop)
R → Rain sensor (1 if raining)

Output

D → Motorized door (0 or 1)

D = 1 if
L = 0 and M = 1
R = 1 and M = 1 and L = 1

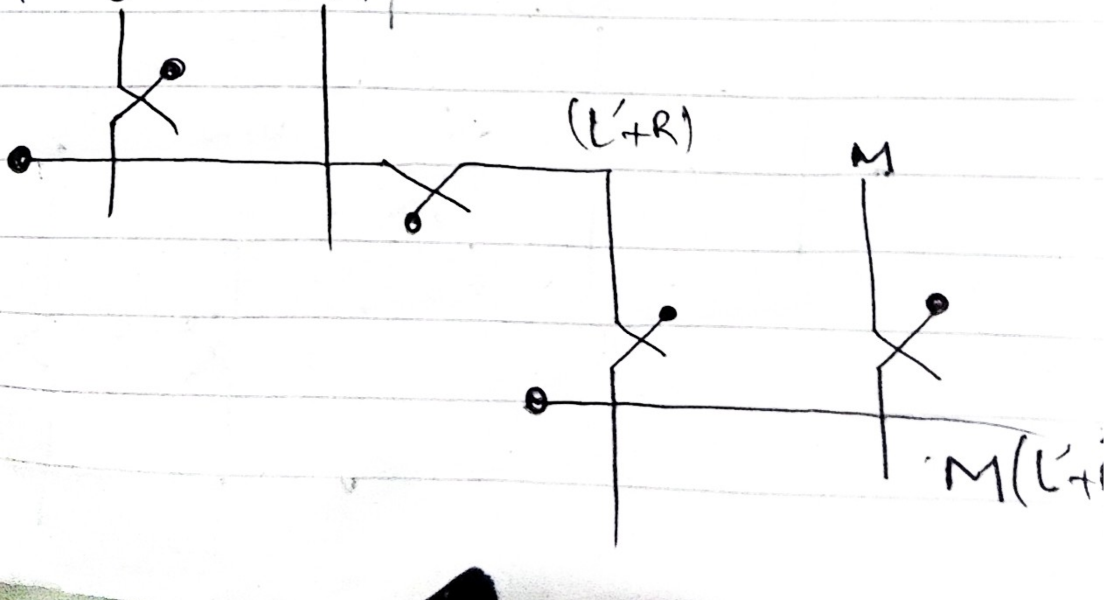
L	M	R	D
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Equation

$D = M \cdot L' + M \cdot R \cdot L$

or

$D = M \cdot (L' + R)$



Q10

Input

B = Battery level (1 if < 15%)

D = Distance (1 if > 2 km)

W = wind speed (1 if > 20 km/h)

Output

S = Saver mode (1 or 0)

S = 1 if

B = 1 and D = 1

W = 1 & B = 1

B	D	W	S
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Equation

$S = B \cdot D + B \cdot W$

or

$S = B(D + W)$